

Lecture 1: Coulomb's Law

Pre-Lecture Worksheet

Complete the following worksheet in preparation for the lecture on Coulomb's Law. The following questions correspond to material covered in chapter 21. Feel free to read chapter 21 and answer the questions. You may also seek answers to the questions from other sources. Note: the textbook will be used as the final source for these questions. While answers gleaned from other sources may be correct, answers must encompass the material covered in the book to be considered correct.

- 1) What are the units of electric charge?

Coulomb

- 2) How many protons are in one SI unit?

$6.241 \cdot 10^{18}$

- 3) What is the charge of one proton?

$1.6 \cdot 10^{-19} \text{ C}$

- 4) Define charge in terms of current and time.

Charge is the product of current and time.

- 5) What is the typical range seen when discussing electric charge?

typically found in the range of 10^{-6} to 10^{-9} C

- 6) True or false: Charges can be created and destroyed?

False

- 7) True or false: Charges flow from an area of low concentration to an area of high concentration.

False

- 8) True or false: Charges flow through conductors.

true

- 9) True or false: Charges flow through insulators.

False

- 10) Provide 2 examples of common conductors.

water, copper

- 11) Provide 2 examples of common insulators.

Rubber, glass

For Questions 12 and 13, circle the noun that makes the statement correct.

- 12) Charging by (conduction/induction) is charging of an uncharged material through physical contact.

- 13) Charging by (conduction/induction) is charging of an uncharged material without coming into contact with the material.